

G. NARAYANAMMA INSTITUTE OF TECHNOLOGY AND SCIENCE

(AUTONOMOUS)

(For Women)

Shaikpet, Hyderabad - 500 104

Department of Computer Science and Engineering

B.Tech Minor in Data Science

III B.Tech. I Semester – External Examinations November- 2025

Academic Year 2025-2026

Scheme of Evaluation

SUBJECT: Introduction to Data Science

SUBJECT CODE: R22-U3111

BRANCH: IT, ETM

Date: 29-11-2025 (FN)

Max Marks: 60

Time: 3Hrs

PART–A

1a) Define Overfitting.

Overfitting occurs when a model learns the training data too well—including noise—resulting in poor performance on unseen data.

1b) Define Data Science and mention any one application of Data Science.

Data Science is the field that uses scientific methods, algorithms, and systems to extract insights from structured and unstructured data. **Application:** Fraud detection in banking.

1c) What is Binary Attribute? Give an example.

A **binary attribute** has only two possible values. **Example:** Gender (Male/Female), Yes/No.

d) Define Mean and Median.

Mean: The average value of a dataset.

Median: The middle value when data is arranged in order.

e) What is a Vector? Give an example.

A **vector** in R is a one-dimensional array containing elements of the same data type.

Example: c(1, 2, 3, 4)

f) What are data frames?

A **data frame** is a two-dimensional table-like structure in R where each column can contain different data types.

g) Write the logical operators in R Language.

&, |, !, &&, ||

h) What is the use of Vectors in R Language?

Vectors store sequences of elements of the same type and are the foundation of R data structures.

i) What is Parametric Data Reduction?

It reduces data by assuming a statistical model and estimating parameters (e.g., using regression coefficients). A model is used to estimate the data, so that typically only the data parameters need to be stored, instead of the actual data. (Outliers may also be stored.) Regression and log-linear models are examples.

j) Define Data Cube Aggregation.

It is a summarization technique in data warehousing where data is aggregated across multiple dimensions to form a data cube.

PART-B

Q2(a) What is Datafication and why is it the future of Business?

Datafication is the transformation of processes, activities, and objects into data that can be analyzed.

Why it is the future of business:

1. Enables data-driven decision-making
2. Improves customer personalization
3. Enhances automation and efficiency
4. Supports predictive analytics
5. Creates new business models
6. Strengthens competitive advantage
7. Helps in risk management

Q2(b) Write a short note on the basic data types of R language.

Basic data types in R:

1. **Numeric** – real numbers (e.g., 3.14)
2. **Integer** – whole numbers (e.g., 5L)
3. **Character** – text strings (e.g., "hello")
4. **Logical** – TRUE/FALSE
5. **Complex** – complex numbers (e.g., 3+2i)
6. **Raw** – raw byte data

Q3(a) Features and applications of R Programming

Features:

1. Open-source
2. Supports vectorized operations
3. Strong statistical libraries
4. Data visualization tools

5. Cross-platform support
6. Community support

Applications:

- Statistical analysis
- Machine learning
- Data visualization
- Bioinformatics
- Big data analytics

Q3(b) Probability distribution and its applications

A **probability distribution** describes how probabilities are assigned to different outcomes of a random variable. The height of humans, following a normal distribution—a bell-shaped curve, also called a Gaussian distribution, named after Gauss. Other common shapes have been named after their observers as well (e.g., the Poisson distribution and the Weibull distribution), while other shapes such as Gamma distributions or exponential distributions are named after associated mathematical objects.

Applications:

- Weather forecasting
- Machine learning algorithms

Q4(a) What is an attribute? Distinguish discrete and continuous attributes.

An **attribute** is a property or characteristic of an object.

Discrete Attribute	Continuous Attribute
Finite/countable values	Infinite values
Example: No. Of students	Example: Height, weight

Q4(b) Explain any two basic statistical descriptions of data with example.

1. **Mean:** average value

Example: (2,4,6) → Mean = 4

2. **Standard deviation:** measures spread

Example: Low SD → values close to mean

3. **Median:** mid value in the data. Eg: 2,4,6. The median is 4.

Q5(a) Explain dispersion of data using Standard Deviation.

Standard deviation (SD) measures how far data points deviate from the mean.

- High SD → high variability
- Low SD → low variability

Formula:

Population Standard Deviation	Sample Standard Deviation
$\sigma = \sqrt{\frac{\sum (X_i - \mu)^2}{N}}$	$s = \sqrt{\frac{\sum (X_i - \bar{x})^2}{N-1}}$

- x represents Each Individual Data Point
- μ is the Mean (Average) of the Dataset
- N is the Total Number of Data Points
- $\sum$ = sum over all the values

Q5(b) Asymmetric attributes and their use

Asymmetric attributes treat presence as more important than absence.

Asymmetric binary: outcomes not equally important.

e.g., medical test (positive vs. negative)

Example: "Has disease (yes/no)" – only “yes” matters.

Used in:

- Medical diagnosis
- Market basket analysis
- Fraud detection

Q6(a) “A factor must have at least two levels” – justify.

A factor represents categorical data.

To form a category, at least two labels are needed; otherwise comparison is meaningless.

Q6(b) Define List and Vector. Explain converting list to vector.

List: collection of different types

Vector: collection of same type

List → Vector:

Use `unlist()` function.

Example:

`unlist(list(1,2,3)) → c(1,2,3)`

Q7(a) Difference between factor and factor level. How can you summarize a factor?

Factor	Factor Level
Categorical variable	Distinct categories of factor

<pre>x <-c("female", "male", "male", "female") print(x) gender <-factor(x) print(gender)</pre>	1. Define a vector with example t-shirt sizes <pre>tshirt_data <- c("M", "L", "S", "M", "L", "S", "M")</pre> 2. Convert the vector to an ordered factor <pre>tshirt_factor <- factor(tshirt_data, ordered = TRUE, levels = c("S", "M", "L"))</pre> Print the resulting factor to verify <pre>print(tshirt_factor)</pre>
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Summarize: Using summary(factor_name).

Q7(b) Write short note on creating named lists and merging two lists.

Named list:

```
x <- list(name="Raj", age=20)
```

Merging lists:

```
merged <- c(list1, list2)
```

Q8(a) Recursive function in R to find sum of N elements

```
sumRecursive <- function(x, n){
  if(n == 0) return(0)
  return(x[n] + sumRecursive(x, n-1))
}
```

Q8(b) Usage of relational operators in R.

Relational operators: ==, !=, <, >, <=, >=

Used in:

- Comparisons
- Conditional statements
- Data filtering

Q9(a) Define Vector. Illustrate vector operations in R.

A vector is a sequence of elements of same type.

Example operations:

```
a <- c(1,2,3)
b <- c(4,5,6)
```

```
a + b
a * b
sum(a)
```

Q9(b) Distinguish While loop and For loop with example.

While Loop	For Loop
Runs while condition is TRUE	Iterates over sequence

Example:

```
i <- 1
while(i<=5){ print(i); i=i+1 }

for(j in 1:5){ print(j) }
```

Q10(a) Pixel-oriented visualization techniques.

A simple way to visualize the value of a dimension is to use a pixel where the color of the pixel reflects the dimension's value. For a data set of m dimensions, pixel-oriented techniques create m windows on the screen, one for each dimension. The m dimension values of a record are mapped to m pixels at the corresponding positions in the windows. The colors of the pixels reflect the corresponding values.

Applications: big data visualization.

Q10(b) Data Reduction: Advantages & Drawbacks

Advantages:

- Less storage
- Faster processing
- Removes noise

Drawbacks:

- Possible loss of information
- Might reduce accuracy

Q11(a) Purpose of Principal Component Analysis (PCA).

Principal components analysis (PCA; also called the Karhunen-Loeve, or K-L, method) searches for k n -dimensional orthogonal vectors that can best be used to represent the data, where $k \leq n$. The original data are thus projected onto a much smaller space, resulting in dimensionality reduction.

The basic procedure is as follows:

1. The input data are normalized, so that each attribute falls within the same range. This step helps ensure that attributes with large domains will not dominate attributes with smaller domains.
2. PCA computes k orthonormal vectors that provide a basis for the normalized input data. These are unit vectors that each point in a direction perpendicular to the others. These vectors are referred to as the principal components. The input data are a linear combination of the principal components.
3. The principal components are sorted in order of decreasing “significance” or strength. The principal components essentially serve as a new set of axes for the data,

Q11(b) Wavelet Transform & its applications

The discrete wavelet transform (DWT) is a linear signal processing technique that, when applied to a data vector X , transforms it to a numerically different vector, X_0 , of wavelet coefficients. The two vectors are of the same length

Applications:

- Image compression
- Denoising
- ECG signal analysis
- Earthquake detection